

Density Functional Theory Calculation of the Bulk Modulus of Germanium

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1 Abstract

This report presents a computational process to determine the bulk modulus of germanium. The lattice parameter and unit cell properties were sourced from available data and used as input parameters to execute density functional theory (DFT) calculations. Next, a script was written and executed to identify the significant figures indicative of energy stability. Using this information, convergence tests were run at various plane-wave cutoff energies (ecutwfc) and k-points to achieve stable energy and pressure outputs. A full unit cell relaxation was performed to optimize the lattice parameter, followed by a series of position relaxations at fractional volumes to determine the ground-state volume and energy. Finally, a Birch-Murnaghan Equation of State was fitted to the energy-volume data and yielded an experimental bulk modulus of 59 GPa. Comparison of this result to other published computational data and relative error are discussed.

2 Introduction

The bulk modulus is the ability of a material to withstand changes in volume when pressure is uniformly applied [1]. Physically, it is equal to the applied pressure divided by the relative change in volume.

$$B = -V \frac{dP}{dV} \quad (1)$$

For the purposes of this computational exercise, we examined the bulk modulus as a function of energy and volume. Substituting $P = -\partial E / \partial V$ into equation 1, we can express the bulk modulus as

$$B = V_0 \frac{\partial^2 E}{\partial V^2} \quad (2)$$

where V_0 is the relaxed volume of the crystal at its ground state energy.

3 Methods

3.1 Crystal Structure and Lattice Parameters

The crystal structure and lattice parameters were sourced from The Materials Project. Germanium has a diamond cubic crystal structure with a two-atom basis, and the lattice parameter for the conventional unit cell are (5.67, 5.67, 5.67) Å [2]. The input file requires the lattice parameter in the unit of Bohr, which converts to 10.87 using the conversion factor of 1.89 Bohr/Å.

3.2 Create Work Directory

A work directory was created to compile the batch scripts needed to perform the computations. The pseudopotential for germanium, input file, Slurm submission script (sbatch), and the executable program for performing DFT calculations (pw.x) were all saved directly from the professor's directory.

```
[jennifer_stefanov@wendian001 Ge_SubmissionFiles]$ ls  
ge_pbe_v1.4.uspp.F.UPF  Ge.vc-relax.in  pw.x  sbatch.sh
```

Figure 3.1 – Working directory elements.

3.3 Edit Input File

The input file provides instructions to the pw.x program for calculation type, system definition, and computational parameters. To define the system for germanium, the input file was edited to:

- prefix = 'Ge', defining the element as germanium
- celldm = 10.87, the unit cell parameter in units of Bohr
- nat = 2, the number of atoms in the primitive unit cell
- ecutwfc = 110, the maximum energy plane waves are allowed to have; this is changed

- iteratively in a later step as part of a convergence test
- nbnd = 16, the number of electron bands to calculate
- ATOMIC_SPECIES = Ge 72.630 ge_pbe_v1.4.uspp.F.UPF, defining the element, atomic weight, and pseudopotential file name
- CELL_PARAMETERS = [0.0 0.5 0.5], [0.5 0.0 0.5], [0.5 0.5 0.0]
- ATOMIC_POSITIONS = [0.00 0.00 0.00], [0.25 0.25 0.25], the fractional coordinates of the atoms in the unit cell
- K_POINTS = 12 12 12, the number of points in a periodic mesh where the computer measures the electrons in a crystal structure; this is changed iteratively in a later step as part of a convergence test

Each of these inputs was sourced from The Materials Project, some of which required unit conversion or scaling. As discussed in section 3.1, celldm units were converted from Å to Bohr, and the nbnd input represents germanium's total number of electrons divided by 2.

```
[jennifer_stefanov@wendian001 input]$ more Ge.scf.110_12.in
&control
  prefix='Ge'
  calculation = 'scf'
  restart_mode = 'from_scratch' !!can also be 'restart'
  tstress = .true. !!calculate stress
  tprnfor = .true. !!calculate forces
  outdir = './' !!same folder is ./
  pseudo_dir = './' !!
  etot_conv_thr = 1.00-4 !!currently placeholders
  forc_conv_thr = 1.00-3 !!currently placeholders
/
&system
  ibrav = 0, !! freeform structure input. In course keep to 0.
  celldm(1) = 10.87, !! lattice constant expressed in Bohr units; maybe convert all to
  Angstrom including below NEED TO UPDATE
  nat = 2, !!number of atoms in primitive unit cell
  ntyp = 1, !!types of atoms
  ecutwfc = 110.0, !!
  nbnd = 16, !!number of bands, how many states for e- am I going to calculate; need 1
  0 for 20 electrons; read from pseudopotential file; NUMBER OF VALENCE ELECTRONS TIMES NU
  MBER OF PRIM CELL ATOMS +30% FOR MARGIN
  occupations = 'smearing',
  degauss=0.002
/
&electrons
/
ATOMIC_SPECIES
Ge 72.630 ge_pbe_v1.4.uspp.F.UPF !!update

CELL_PARAMETERS alat !!volume matrix
0.0 0.5 0.5
0.5 0.0 0.5
0.5 0.5 0.0

ATOMIC_POSITIONS alat !!update coordinates of atoms
Ge 0.00 0.00 0.00
Ge 0.25 0.25 0.25

K_POINTS automatic
12 12 12 0 0 0
```

Figure 3.2 – Sample input file for an scf calculation.

3.4 Evaluate Magnitude of Precision for ΔE

As a practical convergence test, a short script was written and executed to determine the threshold for acceptable deviations of ΔE . The equation is represented by

$$\Delta E = E_{\text{Tot}}V_0 - E_{\text{Tot}}((0.99)^3V_0) \quad (3)$$

$$= \frac{1}{2} \frac{B}{V_0} (\Delta V)^2 \quad (4)$$

where $B = 58$ GPa and $V_0 = 45.69 \text{ Å}^3$ [2].

The output of the script gives a value of 0.007, meaning that convergence is achieved when the

fourth decimal place remains constant between runs of variable ecutwfc and k-point values.

```
File Edit Options Buffers Tools Python Help
# Constants
B0_GPa = 58
V0_Ang3 = 45.69 # Primitive cell V0 for Ge atom from Materials Project
delta = (1.0-(0.99**3))

# Conversion factor GPa * Ang^3 to eV
conv = 0.0062415

# Delta E = (B0 * V0 / 2) * delta^2
delta_E_eV = (B0_GPa * V0_Ang3 / 2) * (delta**2) * conv

print(f"delta_E_eV={}")

[jennifer_stefanov@wendian001 scripts]$ python deltaE.py
delta_E_eV=0.007295419578757339
```

Figure 3.3 – Script for practical convergence test.

3.5 Analyze Convergence of Total Energy and Pressure

Multiple runs were prepared with various combinations of ecutwfc and k-point values to determine which values result in convergence. The ecutwfc values ranged from 60-130 Ry in increments of 10, k-point values ranged from 8-18 points in increments of 2, and the calculation was run as a self-consistent field (scf) to compute ground state properties. Upon completion of the runs, total energy and stress were retrieved from the output files and analyzed for energy stability up to four decimal places. Pressure was analyzed to be less than the order of 10^2 . Based on the chart in Figure 3.5, convergence was achieved with ecutwfc = 110 Ry and k-points = 12.

		60	70	80	90	100	110	120	130
888	ET (Ry)	-427.3014018	-427.3015657	-427.3016266	-427.3018584	-427.3024392	-427.3028078	-427.3028529	-427.3028999
	P (kbar)	5.34	5.08	5.44	4.22	3.84	4.88	5.52	5.03
101010	ET (Ry)	-427.3019663	-427.3021306	-427.3021913	-427.3024228	-427.3030047	-427.3033724	-427.3034177	-427.3034645
	P (kbar)	5.33	5.04	5.34	4.12	3.77	5.00	5.52	5.06
121212	ET (Ry)	-427.3021337	-427.302298	-427.3023587	-427.3025903	-427.3031725	-427.3035398	-427.3035851	-427.303632
	P (kbar)	5.15	5.13	5.36	4.10	3.77	4.93	5.49	4.96
141414	ET (Ry)	-427.3021918	-427.3023561	-427.3024168	-427.3026488	-427.3032304	-427.3035979	-427.3036433	-427.3036901
	P (kbar)	5.29	5.18	5.42	4.25	3.86	4.97	5.62	5.02
161616	ET (Ry)						-427.3036205	-427.30366579	-427.3037125
	P (kbar)						5.06	5.55	5.03
181818	ET (Ry)						-427.3036299	-427.3036752	-427.303722
	P (kbar)						5.00	5.61	5.03

Figure 3.4 – Convergence table for ecutwfc and k-point parameters.

3.6 Fully Relax the Structure

Using the converged ecutwfc and k-point parameters, a complete relaxation of the volume, shape, and atomic positions was run. Updates to the input file were

- Calculation = vc-relax, a variable cell method that allows the cell volume, shape, and atomic positions to change
- ecutwfc = 110, which was selected through the convergence test in the previous step
- wmass = 145.26, the atomic mass of the unit cell
- cell_dofree = 'all', which allows all axes and angles to move
- K_POINTS = 12 12 12, which was selected through the convergence test in the previous step

From the output file, the energy, pressure, and new atomic positions were extracted and analyzed. Of particular interest were the relaxed atomic positions, which expanded from 0.25 in fractional coordinates to 0.250214038. Translating these figures, the lattice parameter relaxed from 10.87 to 10.88 Bohr, or 5.75 to 5.76 Å.

Experimental data has reported the relaxed lattice parameter of Germanium as 5.67 Å [2], and 5.78 Å [3]. Comparing the data presented in this report with other experimental data, the relative error is 0.34%, which is within reasonable limits.

3.7 Compute the Total Energy for the System with Scaled Lattice Parameters

After the full relaxation, position relaxations were performed using fractional volumes to find the best possible V_0 . In the new input file, the cell parameters and atomic positions were updated per section 3.6. The celldm in each run was updated to a fractional lattice parameter, ranging from 0.95 to 1.05 in increments of 0.05. Lastly, the cell_dofree parameter was set to 'shape' to allow only the atomic positions to change, keeping the volume constant. From the output file, the energy and volume were recorded and documented to create a plot of $E(V)$.

	0.95	0.96	0.97	0.98	0.99	1	1.01	1.02	1.03	1.04	1.05
celldm (bohr)	10.3265	10.4352	10.5439	10.6526	10.7613	10.87	10.9787	11.0874	11.1961	11.3048	11.4138
celldm (angstrom)	5.46455	5.52207	5.57959	5.63711	5.69463	5.75215	5.80968	5.86720	5.92472	5.98224	6.03992
E (Ry)	-427.28556	-427.29246	-427.29752	-427.30094	-427.30290	-427.30355	-427.30303	-427.30146	-427.29899	-427.29571	-427.29169
V (bohr ³)	276.00340	284.81140	293.80480	302.98550	312.35550	321.09140	331.67120	341.62070	351.76720	362.11260	372.68830
V (angstrom ³)	40.89944	42.20464	43.53733	44.89777	46.28625	47.58078	49.14854	50.62290	52.12646	53.65949	55.22664

Figure 3.5 – Scaling factors of celldm and associated energy (Ry) and volume (Bohr and Å).

3.8 Fit a Birch-Murnaghan Equation of State

The data from section 3.7 was plotted as a function of energy (Ry) versus volume ($\text{\AA}^3$) and fit to a Birch-Murnaghan Equation of State, which describes the energy of a solid as a function of volume. In modern codes, the fit is modeled as a third-order polynomial substituting the variable $t = 1/\sqrt[3]{V}$ into $E(V) = c_0 + c_1t + c_2t^2 + c_3t^3$. This variable substitution allows us to compute the bulk modulus as

$$B = \frac{t_0^5}{9} \frac{\partial^2 E}{\partial V^2} \quad (5)$$

A script was written to compute t_0 and $\partial^2 E / \partial V^2$ from the fitted curve. Converting angstroms to meters and Rydbergs to Joules, the experimental value of germanium's bulk modulus is computed as 59 GPa.

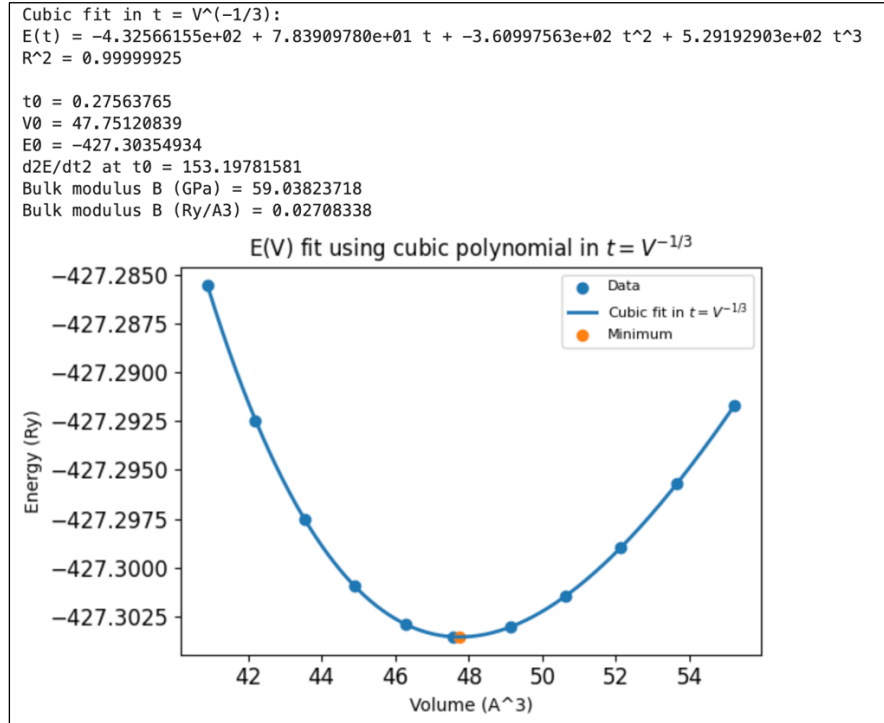


Figure 3.7 – Birch-Murnaghan Equation of State fit of energy versus volume computational data.

4 Results and Discussion

4.1 Bulk Modulus Computation

The computational results of the bulk modulus calculation are in excellent agreement with experimental data. Other sources have reported results of 58 GPa [2] [4], and 63 GPa [3]. When compared to the bulk modulus of 59 GPa in this report, the relative error is 0.84%.

Other results for the bulk modulus of germanium are published as 76 GPa[3] [5] and 80 GPa [6], which

would significantly increase the relative error of these findings to 18.82%. However, these experiments used Local Density Approximation (LDA) instead of Perdew-Burke-Ernzerhof (PBE) functionals to calculate electron exchange correlation, which accounts for the discrepancy. Differences in exchange correlation functions exist because LDA typically underestimates lattice constants while PBE tends to overestimate them, which leads to corresponding differences in bulk modulus values [7].

4.2 Convergence of Pressure

The pressure did not converge to first-digit stability, an oversight that was not noticed until preparing this report. Instead, the goal was interpreted for the pressure to be less than 10 kbar outright. Due to the strong agreement between the bulk modulus computations in this exercise and other experimental data, the effect of this error is likely insignificant. However, it would be beneficial to extend *ecutwfc* and *k*-point convergence tests past 110 Ry and 12 points to see if the pressure eventually becomes first-digit stable.

5 Conclusion

In this work, the structural and mechanical properties of germanium were investigated using DFT computations. A convergence test was executed to control for meaningful precision, and the unit cell relaxed to optimize the lattice parameter and determine the unit cell volume. Using these figures, the bulk modulus was calculated by fitting energy–volume data to the Birch-Murnaghan Equation of State.

The calculated equilibrium lattice constant, 5.75 Å, was found to be in excellent agreement with previously reported experimental values for germanium, resulting in a relative error of 0.34%. Similarly, the calculated bulk modulus of 59 GPa was consistent with reported values, representing a relative error of 0.84%. Discrepancies between experimental values may be due to the exchange correlation functional used in the calculation, as well as factors such as *k*-point sampling, plane-wave cutoff energy, and convergence thresholds.

6 References

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